THEO TRUSS

Architecture

PORTFOLIO: 2023-2026

I

ACCRETION

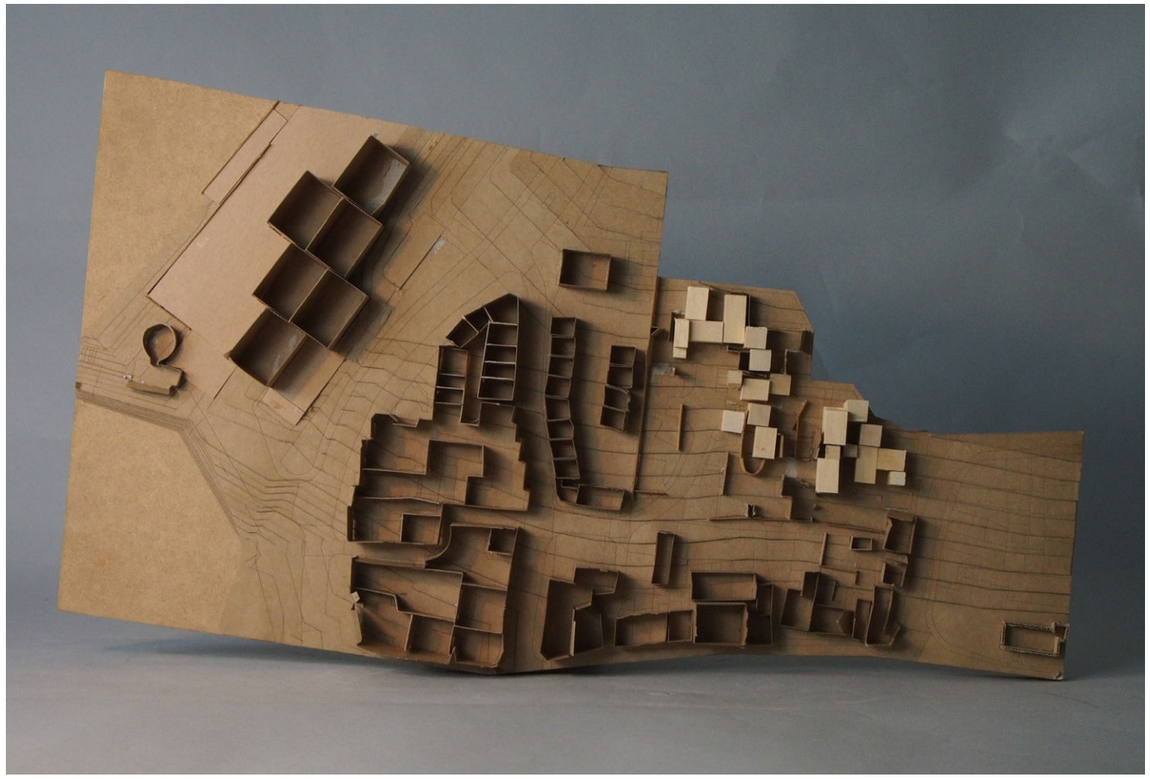
Margate, Kent

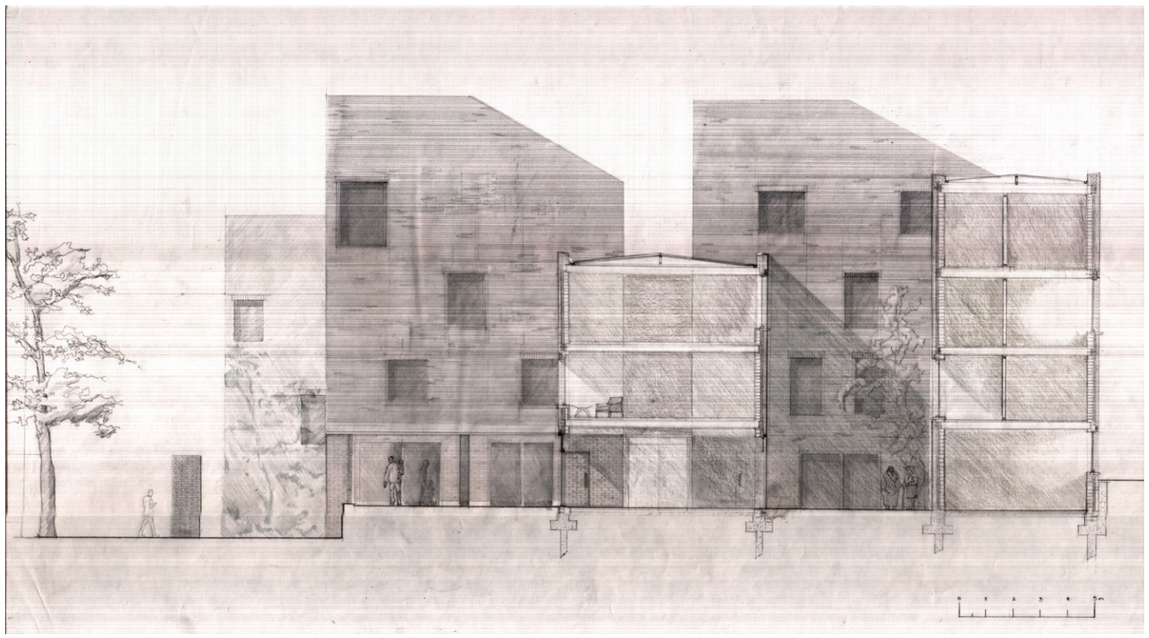
The topography, natural and urban, of the Old Town of Margate, has undergone various eras of change since its growth from a small settlement in the late middle ages. Whereas the early accretion of the Old Town grew from key routes to the sea, the mid 18th century saw the introduction of squares and other internally facing developments, refacing and often covering up original timber framed structures. In the early 19th century, much larger structures such as Cobbs Brewery started to interrupt the scale of the spaces of the old town. Today, many of these structures are replaced by equally large modern iterations, such as a care-home that now sits over the site of the brewery.

This project learns from previous structures and contexts, mapping them not to recreate the past, but instead to learn from the qualities of spaces that used to exist, to create a new iteration of this topography, dissolving oversized structures down to the scale of the Old Town. By altering the existing concrete frames, and strategically placing new walls, resilient and sustainable spaces may emerge. This particular iteration uses the spaces for a shared programme: an assisted living complex (which provides a variety of accommodation from more independent units, to ones that require closer support) that shares an urban block interior with a primary school and nursery. The groundline of the site, moving up the side of the valley that Margate sits in, provides opportunity for these functions to operate more independently.

Similar to the previous changes of character of the buildings in Margate, the existing frames and the new frames of the buildings would be refaces in a dark, striated material such as a dark brick, mixed with reclaimed material from local demolitions, such as slate. In this way, the structures will seem to rise from the ground, be of the earth, as well as becoming a stratigraphic representation of built memory. Within the urban block perimeter, a path is drawn from King St (at the bottom of this drawing), walking up the valley towards the sea, drawing a walkable route through this block of the Old Town, where before it was impassable.

Block shadow plan.
1:250, A1 ▶



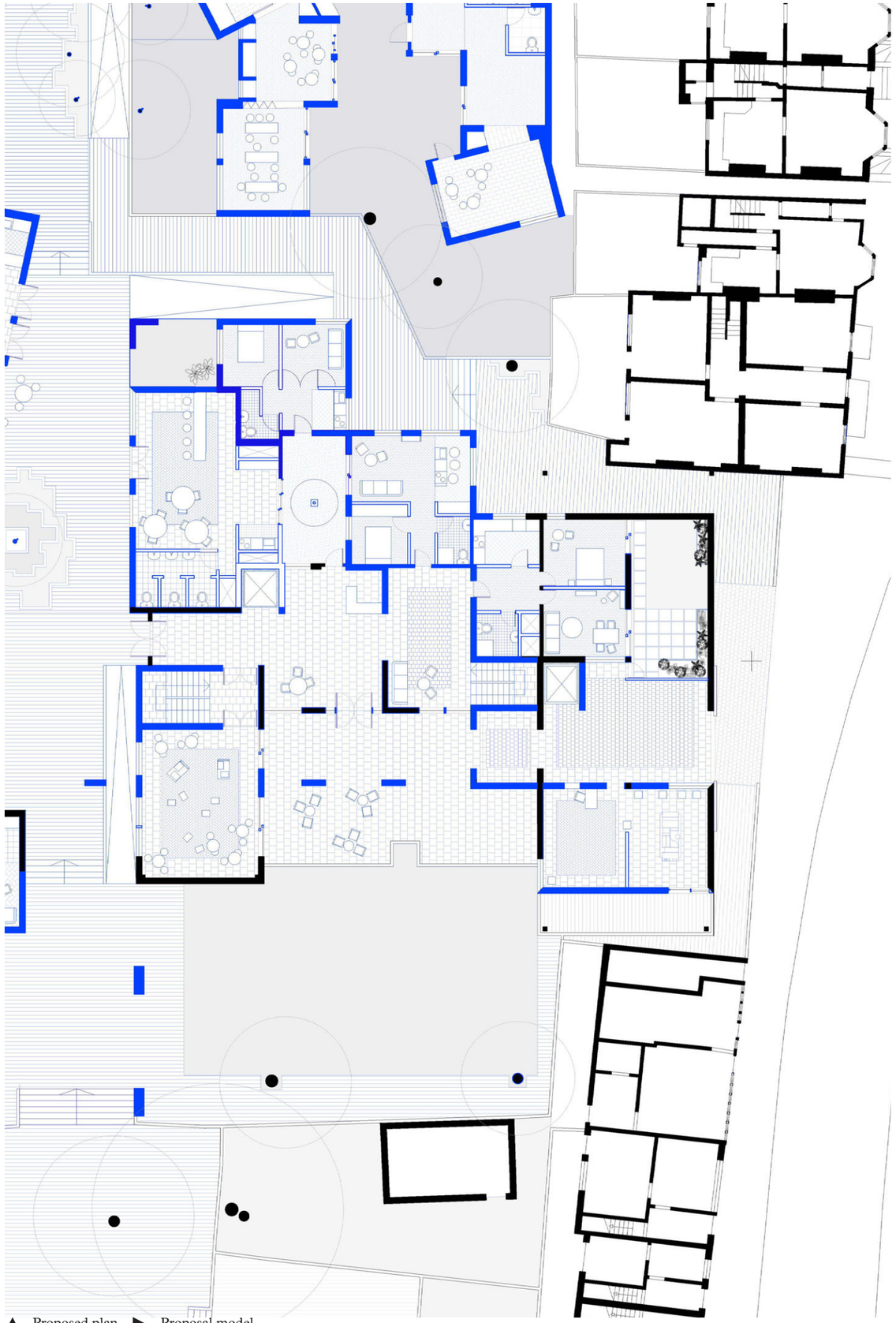


The cellular arrangement of the buildings allow a variety of spaces to emerge. Precedents include Aldo van Eyck's Amsterdam Orphanage, Hans Scharoun's Darmstadt Primary School, and the Antonianiko Complex in Vatheia, Greece. In this way, the clusters are resilient enough to have internal and external spaces that support varying uses and degrees of privacy. Ultimately these volumes are to be wholly of the landscape.

▲▲ Proposal context model 1:500

▲ Assisted living cluster short section, 1:50

▶ Proposal cluster model 1:100



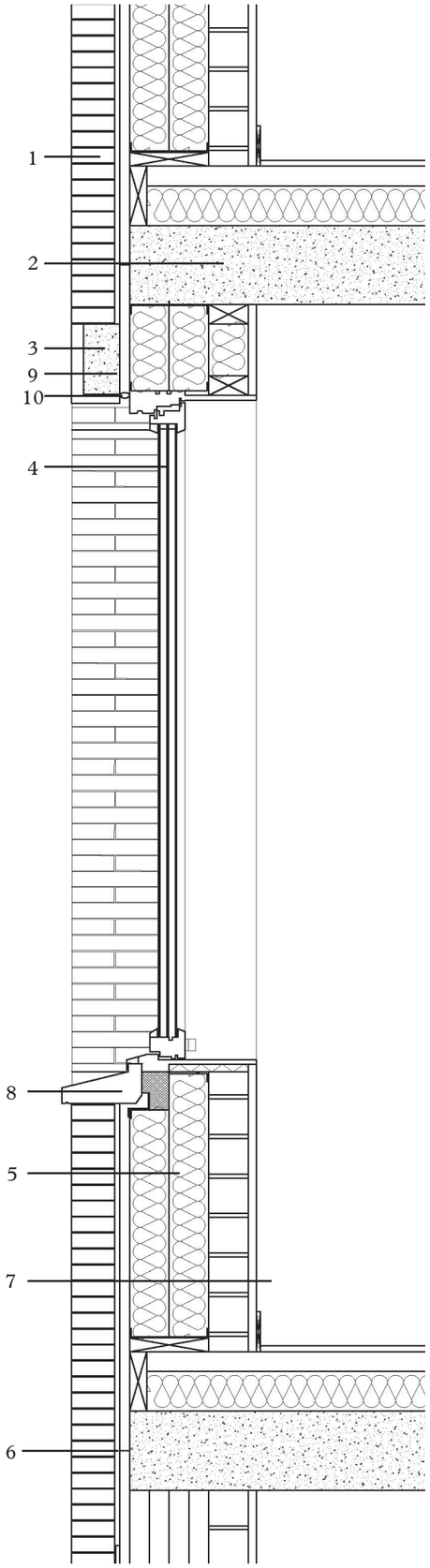
▲ Proposed plan
1:100, A0

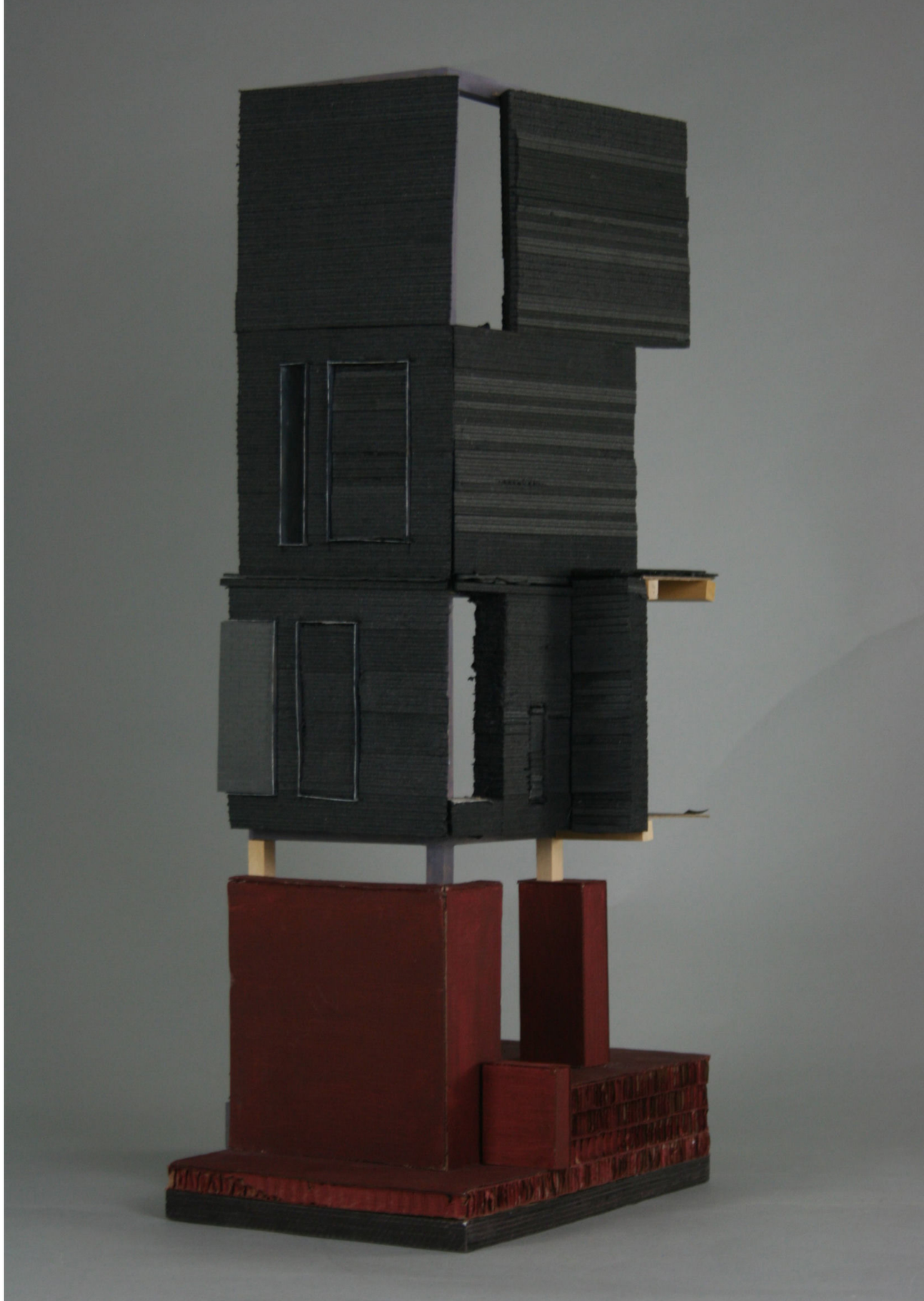
▶ Proposal model
1:250

The cellular clusters also adapt over time, as the same key walls can easily accommodate multiple cells being joined together, for example if two single apartments were joined together. Where the existing floor slabs must be cut, the remaining material around each column (approx. 100mm) can give opportunity for another layer of insulation.

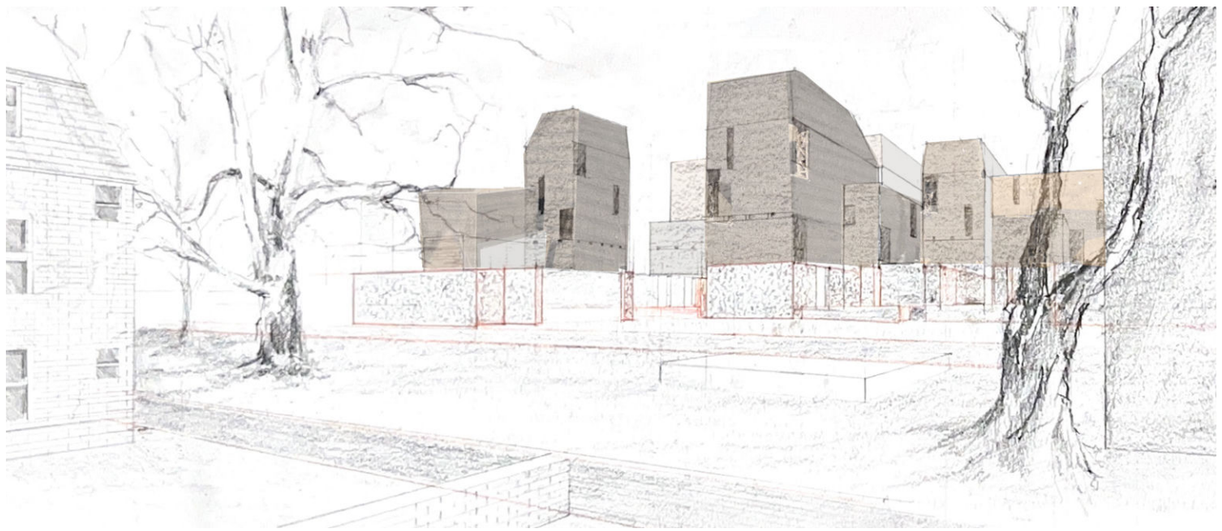
▶ Wall construction section
1:50, A2
⏩ Materiality and structural model
1:20
▼ Nursery concept collage

1. Brick outer leaf, (Petersen Tegl Kolumba Bricks) 109mm. Patches of reclaimed tile and slate from local demolitions are also bedded into the mortar.
2. Existing concrete frame 200mm deep
3. Precast concrete lintel with brick slip facing
4. Triple glazed window, frame set in a Scottish rebate
5. Mineral wool insulation, 200mm, filling the additional 100mm depth left by the cutting of the existing slab.
6. Outer leaf tie
7. Inner blockwork 125mm
8. Precast concrete sill.
9. Vertical DPC
10. Mastic sealant





▲ Extracts from a series of maps that I made, illustrating the growth of Margate up to 1900.
▶ Looking down the proposed scheme, towards King St.
▼ Internal block perspective, looking up the valley.



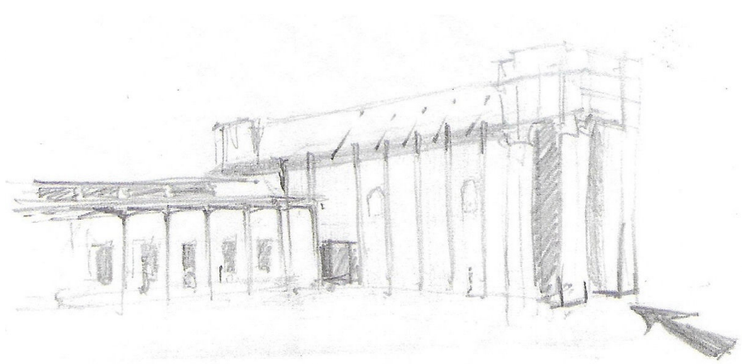
II

ARTEFACTS

The Fitzwilliam Museum, Cambridge

The Fitzwilliam Museum sits as an island, the site and the collections both disconnected from the city of Cambridge. In this second-year project, a masterplan proposes careful alterations and changes to the museum fabric, as well as new routes around Cambridge, that help better integrate the Fitzwilliam collections into the city. Any museum may be considered as a home for the artefacts and the art (crucially an *impermanent* yet significant one). The Fitzwilliam - Greek Revival and distinctly imperial - is built to appear permanent to establish authority, and therefore a perception of authenticity. However, the building currently fails to recognise its relative impermanence to the artefacts that it is responsible for. Much of the challenge therefore is to give power back to the artefacts, and indeed the stories, memories, and people behind them. The museum both as an object, and a protector of history, is inherently associated with the imperial leverage over the acquisition of artefacts in its past. The origins of the initial funding for the museum is rooted in the trans-Atlantic slave trade, and any new intervention to the museum must come to terms with this. The museum must be an instrument to teach a transparency about this, as well as one made to serve the objects and stories within.

This project proposes purposeful interventions to the building fabric. One of the boldest is to place the entrance on the end of the eastern elevation (right), cutting though the southernmost projection of the portico. This encourages the visitor to enter the building before experiencing the tight intercolumniation of the portico, which no longer locks the museum from the city, but instead frames it. Visitors enter the museum into new loggias (built using reclaimed brick and stone from the current steps), with 'in-between' spaces employed as both new exhibition space but also as to de-centre the visitor from the current walls of the museum. The materiality and construction of these loggias draws inspiration from many of Hans Döllgast's projects in Munich.



New Founder's entrance from Fitzwilliam St, Oil on Canvas ▶

Thresholds of the Museum

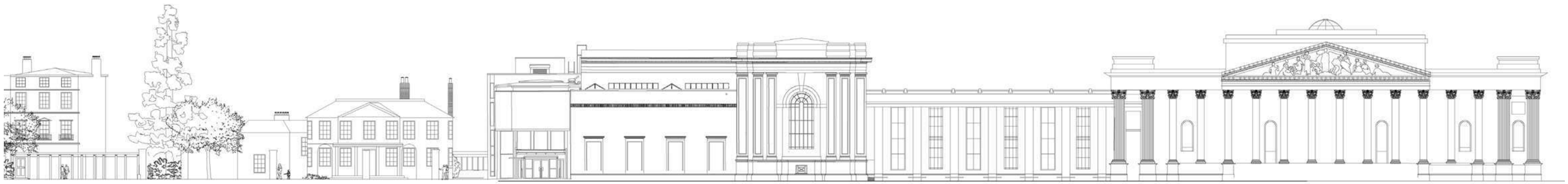
The second incision into the fabric of the museum, through the stairwell of the 1970's extension to the south (below), results in a similar gateway to the Founder's building cut. These tall passageways allow the visitor to confront the material of the museum as an artefact itself, the different layers of which are made legible through more accessible entrances. Where the Founder's building entrance is moved to be step-free, the new entrance for the 1970's extension reorientates the museum to acknowledge its length, also reinstating the courtyard that was filled in by an early 2000's extension. The new St Peter's Terrace entrance, to the south of Grove Lodge (Left), references both the existing portico of the house, as part of the aim to create a more domestic environment for exhibiting artefacts, and also the other gate typologies that are found in the rest of the proposal. For instance, gates made from material reclaimed in the alterations to the existing building fabric would mark trails around Cambridge and the fenland, joining together other museums, and spaces designed to deliver a talk to a small group. Over time, the gates would be designed to decay, leaving only their stone feet.

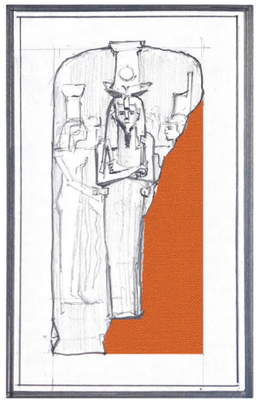
▲ Alterations to the cafe entrance, with Grove Lodge Schoolhouse to the left
1:50
◀ Proposed St Peters Terrace Entrance
1:50

Each of the three interventions help establish a new axis in how the museum is experienced, acknowledging its length while dissolving the harsh boarders to the street.

▲ The three proposed entrances for the Fitzwilliam Museum

▼ Eastern Elevation
1:500, A1



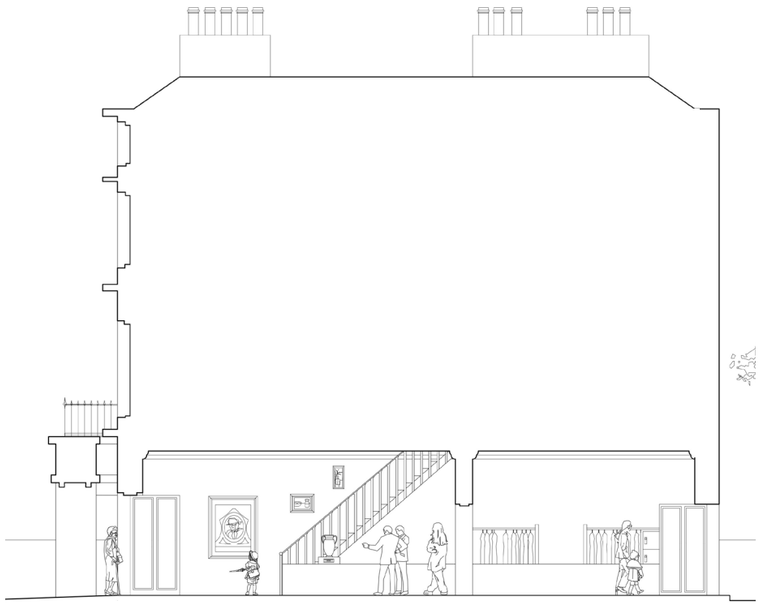


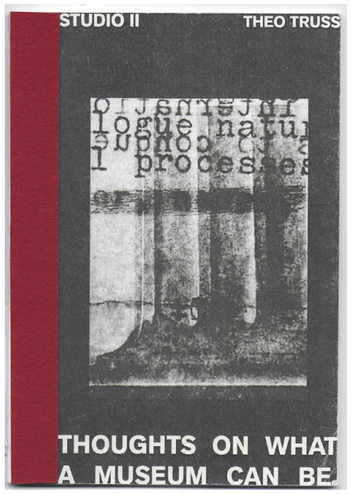
By making spaces where artefacts can be displayed in different ways, for example in the more domestic setting of St Peters terrace, the collections can better integrate with the city. By making space for demonstrations, and by moving artefacts, the museum buildings are more resilient in adapting to the needs of the collection. Much of this project focused on material reuse, and by building a schedule of available materials, new solutions do not have to depend so heavily on new materials. A short book created to accompany this project notes key discussions and ideas behind asking what the Fitzwilliam Museum can be. In establishing a new educational component to the museum, this booklet discusses how teaching spaces can integrate into both the domains of the city, and the museum, as well as showing designs for educational toys that I designed alongside this project, based off Froebelian gifts.

◀ Proposed St Peter's Entrance Model. 1:50.

◀ Artefacts from exploration of ideas. From top to bottom: Slate new axis diagram, drawing of Ramessess III coffin lid, educational alphabet toy prototype

▼ Long section though the proposed St Peter's Terrace entrance. 1:50, A2

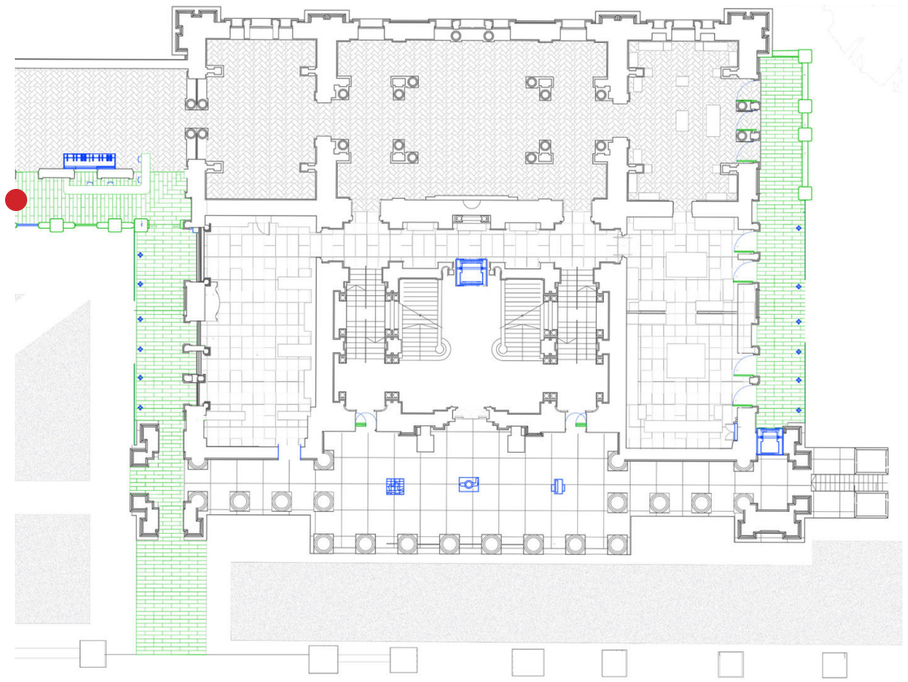




▲ Cover for a project booklet that I made, containing many of the discussions that drove the ideas behind the project,

▶ Inside the closed loggia adjacent to the Founders Building

▼ Proposed plan for Founder's Building Entrance, attempting to show reused material (green) with new material (blue). Perspective in model photo taken from the red dot. 1:200, A1.



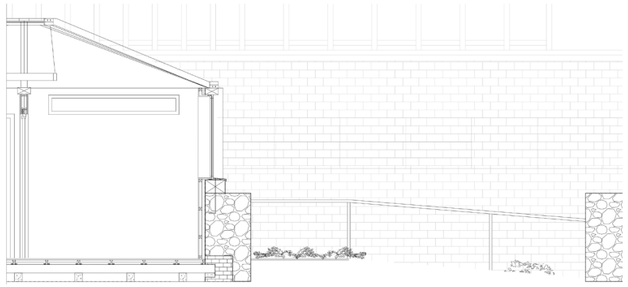
III

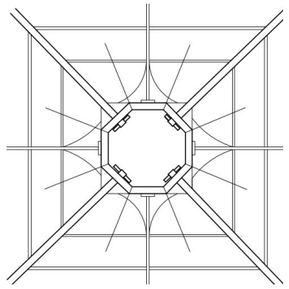
REFLECTION

History Faculty Building, Cambridge

The History Faculty building at the University of Cambridge is currently under renovation. The Grade II* listed building, designed by James Stirling in 1968, needs urgent conservation work. The current proposal also looks to integrate the building into the surrounding Sidgewick site. This projects brief called for a fully demountable structure that sat in the corner of the site, providing a new valuable space for the library and surrounding environment. The project presents a multi-faith reflection space, that nestles in the existing ramparts of the Seeley Library.

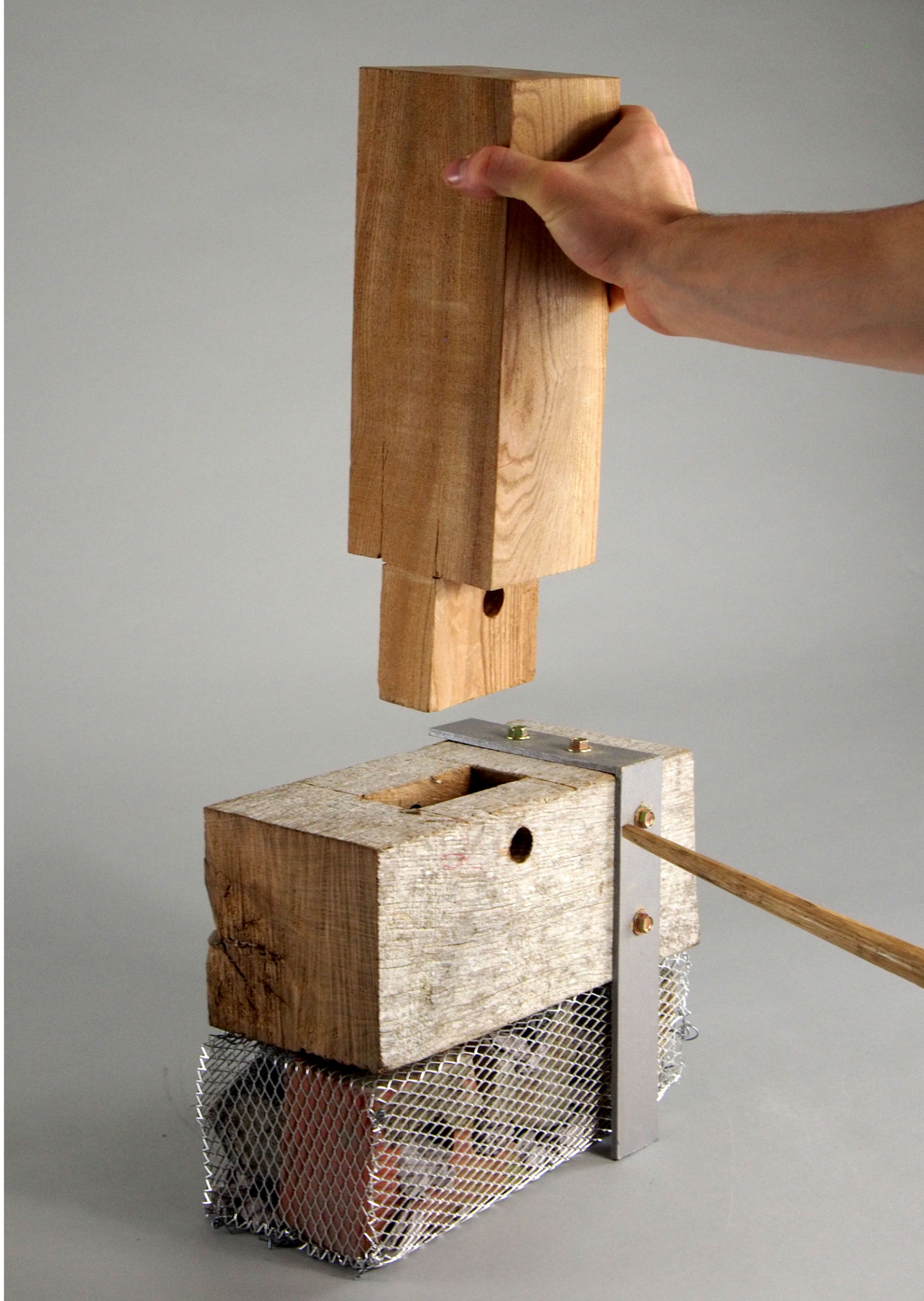
A timber framed construction would sit onto a gabion wall system, being tied down in between timber columns. I investigated possibilities of using pinned mortise and tenon as well as other traditional barn building joinery as to support a roof that it topped by a skylight.





▲ Clockwise: long section through reflection space entrance, roof light detail, model (1:100)

Connection detail model, 1:1 ▶

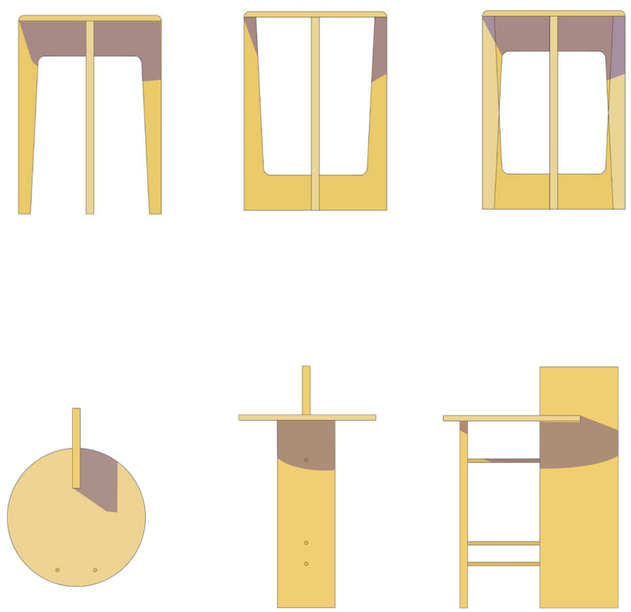


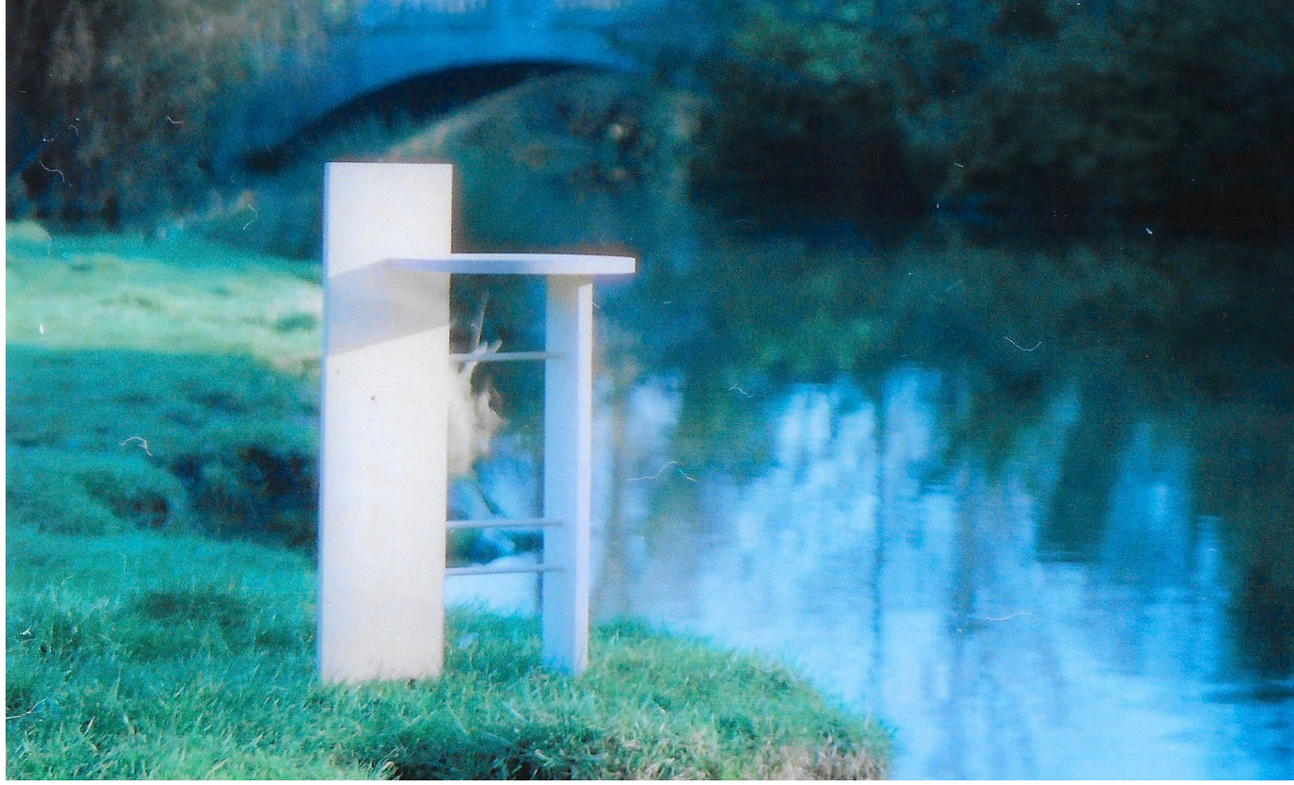
IV

NOVIGO

Novigo is a design and furniture studio that I co-founded with another architecture student from Cambridge. Novigo was created as a way to solve the growing problem of inflexible and unsustainable furniture. By finding creative designs in reclaimed timber, multiple configurations for each piece may be constructed from a kit of parts, reducing waste while still using quality timber. As Co-Founder of Novigo, I learned a great deal from enterprise competitions, aiming to turn Novigo into a social enterprise that would use the talent of local designers and provide short term jobs. Although in its infancy, our ideas have been awarded seed funding by one competition, and a runners-up prize in another.

The prototyping and design process has been a wonderful challenge, and has developed my carpentry in thinking about a product design. These two prototypes use no glue, screws or other fixtures, and rely on simple joints and carpentry that is designed to be easy to teach.





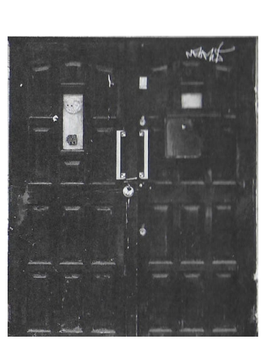
NOVIGO

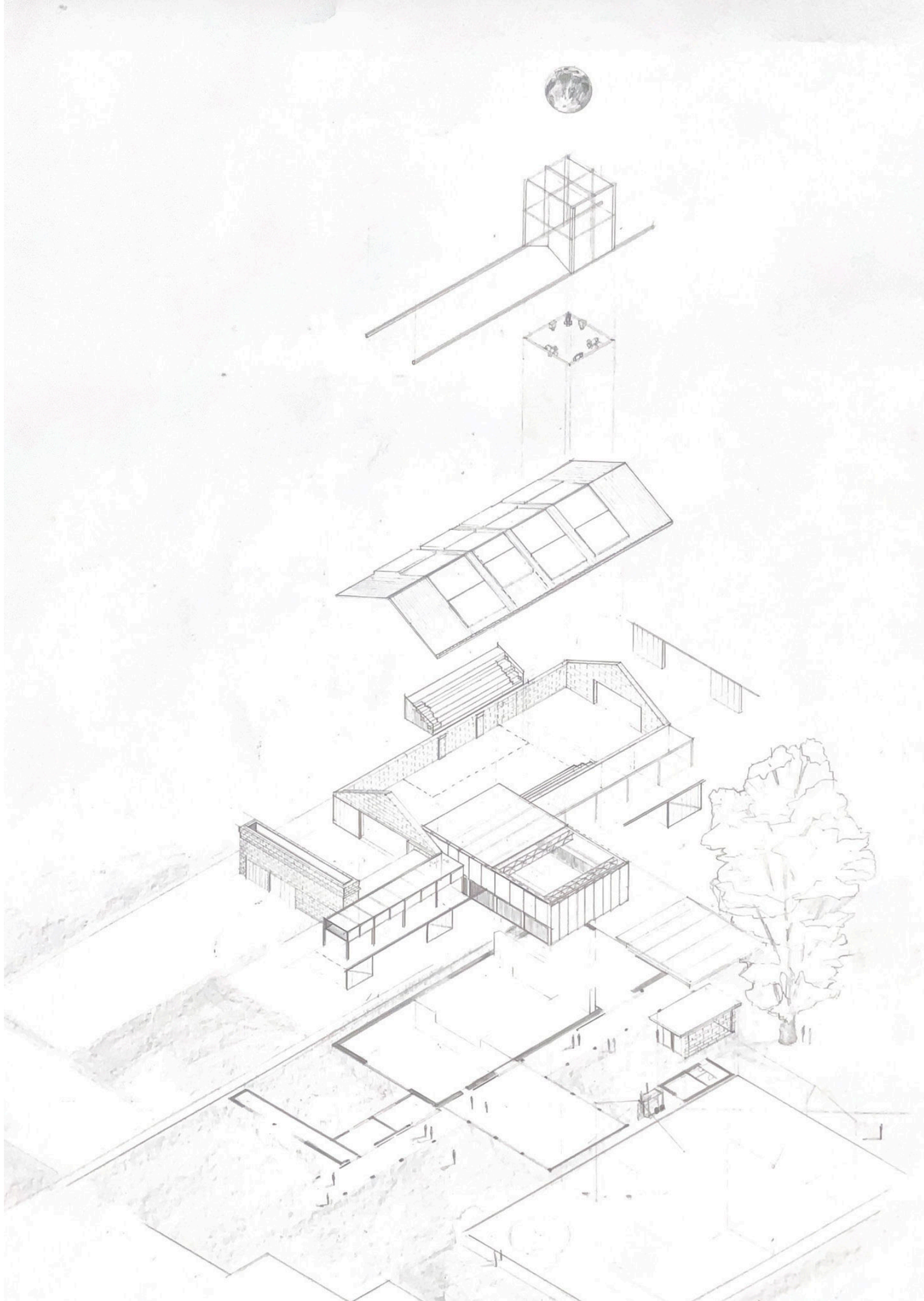
V

A STAGE, BACKSTAGE

Mill Road, Cambridge

This first-year project proposes a new theatre on Mill Road, a community with a strong sense of identity, that holds firm amongst the changing environment and increased inequality in Cambridge. The design was initially guided by Thornton Wilder's *Our Town,* a play which deals with themes of community and loss, and is written with very little set, the stage manager playing a key role in the play itself. This inspired my proposal, using the mechanics of the barn typologies as a theatrical device to reveal, or obscure views to the town outside. The audience sits in a transverse stage, facing other audience members as well as the play. A moveable lantern in the roof of the former pool hall also acts as the fly tower, moving from opening to opening between each act. The aim of the theatre was to construct spaces for the community of Mill Road, a venue where performances include the town itself.





VI

RESEARCH

Spomenici (former Yugoslavian monuments) have been previously examined through historical, political, and popular lenses, yet have not been systematically examined architecturally. Existing frameworks do not allow analysis of the monuments' material lives with embodied spatial experience, often leaving them as works of art at an architectural scale rather than active spaces of memory. This dissertation introduces the concept of the memorial horizon, or the overlap of ontological registers: ideological, civic, chthonic and phenomenological, revealing where they become legible in the monuments' material, using architectural analysis to reveal the mechanics of memory politics in ways that can supplement other disciplines. Through fieldwork across 5 former Yugoslavian states, I selected case studies in order to demonstrate how material fractures in monuments make legible the layered contested histories of post-socialist memorialisation.

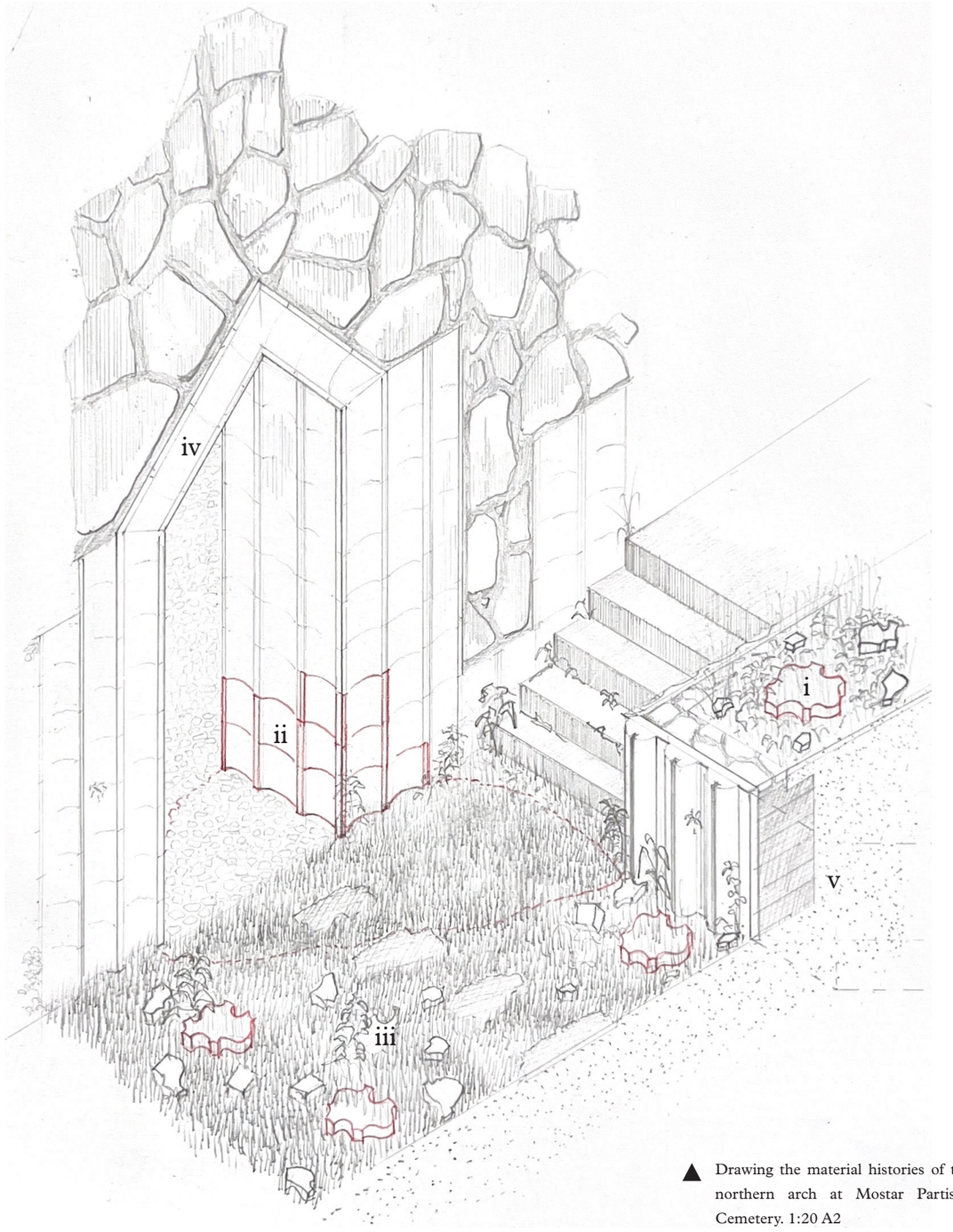
i. Gravestones. Many destroyed between 1992-1995. Restored 2018. Over 700 destroyed in 2022.

ii. Areas damaged by 1992 bomb damage

iii. Wild plants: *Mentha longifolia* (Mint), *Sempervivum tectorum, Taraxacum officinale* (Dandellion)

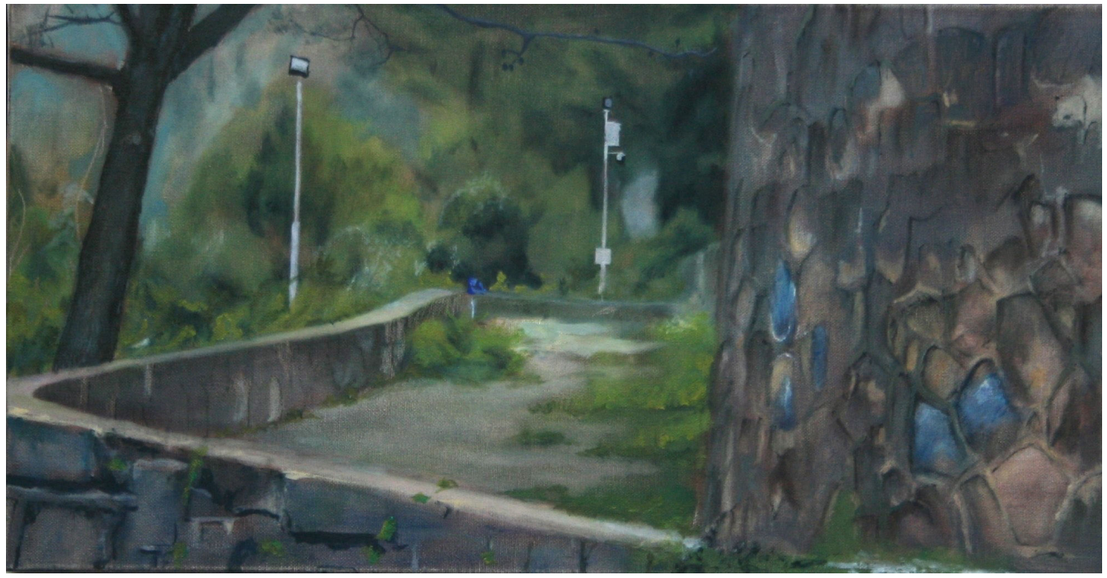
iv. Limestone cladding elements used to frame the arches

v. Retaining walls, clad limestone.



▲ Drawing the material histories of the northern arch at Mostar Partisan Cemetery. 1:20 A2

◀ Surviving gravestone at Mostar. Oil on canvas



▲▲ Ascending the cemetery at Mostar.
Oil on canvas

▲ Fragments of the notebooks from the fieldwork.

◀ Mostar Partisan
Memorial Cemetery

◀◀ Monument to Partisan Munition Factory - 'Bomb Monument', Skopje. Oil on panel.

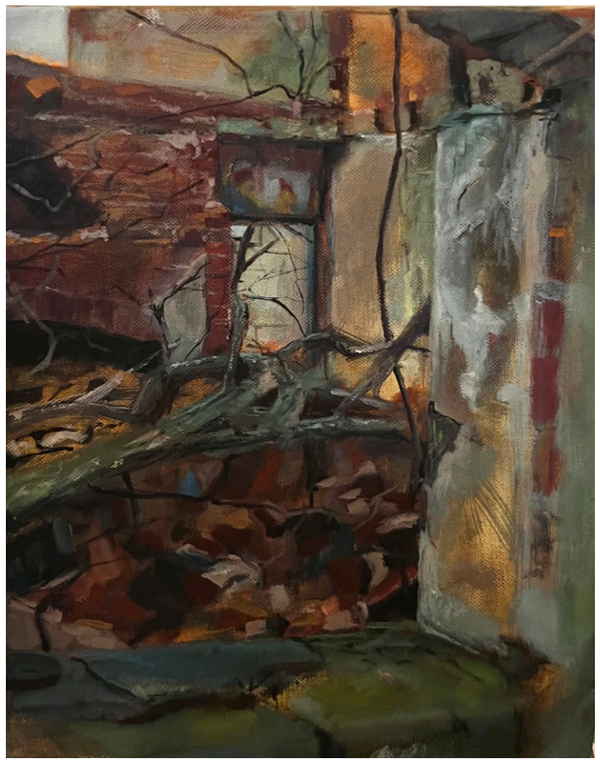
◀ Monument to Women Fighters, Skopje

▲ Bubanj Memorial Park (***Спомен-парк Бубањ***) "The Three Fists" Niš, Serbia

VII
PAINTING

The act of painting - observing and making marks (decisions) about the quality of space - is something that aligns with the way that I communicate in architecture. People may suggest that it is too vague, or slow to work through architectural ideas in this way, but with any sustained project or attention to a place, room, single brick, one learns so much more about its character and purpose. In this portfolio, I have chosen a variety of pieces that show my voice as an artist, as a thorough and thoughtful creator and communicator.

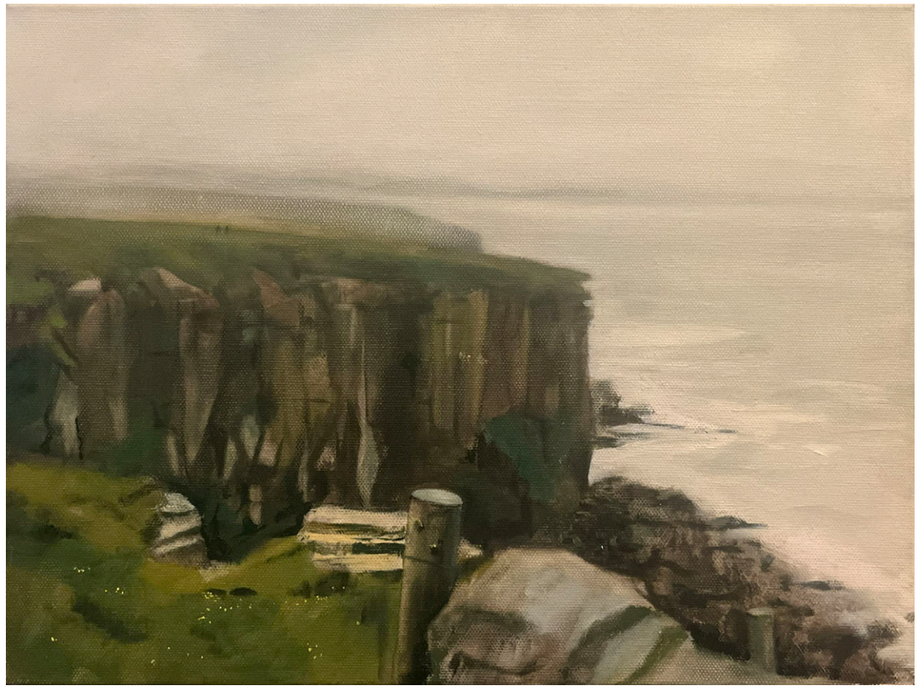
These two pieces for example, are a part of an ongoing project observing Barn Cottage, an abandoned structure a short walk from where I live, losing more of its walls with each storm. Part of the motivation for the project is to see how the shapes change, internally as it fills with bricks, or externally as it is reclaimed. I remember visiting after a particularly bad storm, only to find the silhouette changed. Of course part of the beauty is that it changes, but being honest with myself I was disappointed to see that the chimney had fallen in.



▶ Barn Cottage I, oil on canvas.

◀ Barn Cottage II, after the storm in December 2024. Oil on canvas

Naturally, the focus of many of my paintings is a certain attention to making space, whether that be how space is changed by time, or how space is framed in entrances and openings. I am also fascinated however, by what happens when our understanding of land, in navigation through the grain of a city, or landscape, comes to a border. Often this is in landscape, either at the edge of water, or other lands impassable by foot. There is something about the way we face these spaces in the buildings and demarcations that we create that really fascinates me.



▲ Loch Eriboll, Highlands. 40cm x 40cm, oil on canvas.

◀ At Duncansby Head Lighthouse. Oil on Canvas

◀◀ August garden at Thorame Basse, France. Oil on canvas

◀ Lady Chapel of Little St Mary, Cambridge. Oil on Canvas

▼ Faneromeni Monastery in Lefkada, Greece. Oil on Canvas